

VENKATESH KELAM

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SUMMARY

Software Engineer with 3+ years of experience building scalable distributed systems and AI-powered applications. Expertise in event-driven architectures, high-throughput data pipelines, and cloud-native systems on AWS. Experienced in designing and deploying backend services, optimizing data workflows, and integrating real-time AI capabilities. Consistently shipped high-impact systems across healthcare, real estate, and AI-driven SaaS platforms with a focus on reliability, performance, and scale.

WORK EXPERIENCE

IBM

Software Engineer

Bellevue, WA
Feb 2026 - Present

- Developed a production-grade embeddable chatbot (CaaS) with real-time streaming, multi-turn conversation, and multi-tenant support.
- Architected & delivered a scalable full-stack SaaS platform for AI-driven property management, built on FastAPI, React & PostgreSQL.
- Engineered a scalable geospatial rent comparables engine surfacing nearby market data enriched from diverse external listing sources.
- Implemented ML-based rent forecasting & vector similarity search to power real-time pricing recommendations across 1K+ properties.
- Built an async Celery & Redis pipeline automating daily AppFolio data ingestion across rent rolls, vacancies & financials at scale.

Ipser Labs

Software Engineer Intern

San Jose, CA
Feb 2025 - Jul 2025

- Built Dhorobha, a scalable African real estate marketplace with property listings, advanced search filters & AWS-backed data pipelines.
- Launched an interactive property search platform with real-time map markers, enabling seamless address-based property discovery.
- Optimized DynamoDB data models with secondary indexing, improving read/write throughput and query efficiency at production scale.
- Enforced rate-limiting & API governance across distributed services, cutting invalid traffic by 40% and strengthening backend security.
- Refactored broken CI/CD pipelines using GitHub Actions, streamlining build, test, and deploy workflows and cutting failures by 35%.

Accenture

Software Engineer

Chennai, India
Jul 2021 - Jul 2023

- Developed scalable healthcare applications with responsive React frontends, enabling seamless clinical workflows & real-time visibility.
- Enhanced frontend performance by implementing lazy loading and code splitting, reducing page load times across healthcare modules.
- Optimized complex SQL queries, window functions, and joins, improving data extraction and overall query performance by 25%.
- Engineered REST APIs with FastAPI and Flask, enabling reliable data exchange across enterprise systems and third-party services.
- Boosted database performance by 20% under high concurrency via strategic indexing, query optimization, and connection pooling.

SKILLS

Certification:	AWS Certified Solutions Architect - Associate
Languages:	Python, JavaScript, TypeScript, Java, SQL
AI & ML:	OpenAI API, LangChain, PyTorch, pgvector, RAG, SSE Streaming
Backend:	Django, Flask, FastAPI, Node.js, REST APIs, GraphQL, OpenAPI/Swagger
Databases & Caching:	PostgreSQL, MySQL, Redis, MongoDB, DynamoDB, PostGIS
Cloud & DevOps:	EC2, S3, CloudWatch, CloudFormation, RDS, Lambda, SQS, SNS, EFS, Docker
APIs & Services:	REST/RESTful APIs, GraphQL, Kafka, OAuth

EDUCATION

Northeastern University

MS in Computer Software Engineering, GPA - 3.96 / 4.0

San Jose, California

Anna University

B.Tech in Information Technology, GPA - 8.56 / 10.0

Chennai, India

PROJECT EXPERIENCE

[Real-Time Cloud Data Pipeline](#)

Feb 2023

- Built a serverless AWS pipeline (API Gateway, SQS, Lambda) processing millions of transactions using event-driven architecture.
- Architected async event-driven workflows across API Gateway, SQS, and Lambda, enabling decoupled and scalable data processing.
- Configured CloudWatch rules to transfer data from RDS to Redshift every 15 minutes, enabling real-time analytics and reporting.

[Healixity](#)

Feb 2025

- Developed real-time dashboards for health data streams, reducing API latency by 73% (450ms → 120ms) via query tuning and caching.
- Implemented end-to-end metrics and structured logging to proactively detect and diagnose production performance bottlenecks.
- Optimized backend data access patterns and indexing strategies to significantly improve query efficiency under high-load conditions.